PRACTICAL FRIDAYS

OCC: SOFTWARE ENGINEER

YEAR 2

DESIGN AND MANIPULATE DATABASES

(MDB622)

SAQA QUALIFICATION ID: 119458



CTU training solutions | Private Tertiary Education

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Practical Fridays Goal

Students will design, create, populate, and manipulate SQL Server databases and integrate them with C# desktop applications. Throughout the semester, students will progressively develop database applications while applying industry best practices in database design, normalization, security, and reporting.

Learning Outcomes

By the end of the semester, students should be able to:

- Design Entity Relationship Diagrams (ERDs)
- Apply normalization techniques (1NF, 2NF, 3NF)
- Create SQL Server databases using SSMS
- Create and manage tables, relationships, and constraints
- Perform CRUD operations using SQL and C#
- Implement database connectivity using ADO.NET
- Create stored procedures and views
- Implement authentication and authorization
- Generate reports and dashboards
- Apply transaction management
- Develop complete database-driven business applications

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PHASE 1: FOUNDATION (BEGINNER)

Week 01 (20 – 24 July 2026): Database Design Fundamentals

Activities

- Introduction to SQL Server
- Introduction to SSMS
- Database design principles
- ERD creation
- Database normalization
- Primary & Foreign Keys

Deliverables

- Draw ERDs for:
    - Library Management System
    - Inventory Management System
    - Student Management System

Concepts

- Entities
- Attributes
- Relationships
- Cardinality
- Data Types
- Constraints
- 1NF, 2NF, 3NF

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Week 02 (27 – 31 July 2026): Student Management System

Tables

- Students
- Courses
- Departments

Features

- Add student
- Update student
- Delete student
- Search student
- View student profile

Database Concepts

- CRUD Operations
- Primary Keys
- Foreign Keys
- Relationships
- Basic Queries

C# Concepts

- SQL Connection
- DataGridView
- Forms
- Input Validation

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Week 03 (03 – 07 Aug 2026): Employee Management System

Tables

- Employees
- Departments
- Positions

Features

- Employee registration
- Department assignment
- Employee listing
- Search functionality

Database Concepts

- One-to-Many Relationships
- INNER JOIN
- LEFT JOIN
- UPDATE Queries

C# Concepts

- ComboBoxes
- Search Filters
- Data Binding

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Week 04 (10 – 14 August 2026): Library Management System

Tables

- Books
- Authors
- Members
- Loans

Features

- Borrow books
- Return books
- Book availability
- Member management

Database Concepts

- Composite Relationships
- Referential Integrity
- Date Handling

C# Concepts

- Multiple Forms
- Record Navigation
- Validation Rules

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Week 05 (17 – 21 August 2026): Hotel Reservation System

Tables

- Guests
- Rooms
- Reservations

Features

- Room booking
- Availability checking
- Check-in
- Check-out

Database Concepts

- Date Calculations
- Business Rules
- Validation Constraints

C# Concepts

- DateTimePicker Controls
- Booking Logic
- Search Functionality

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PHASE 2: INTERMEDIATE DATABASE APPLICATIONS

Week 06 (24 – 28 August 2026): Vehicle Rental System

Tables

- Vehicles
- Customers
- Rentals

Features

- Vehicle booking
- Rental history
- Return processing

Concepts

- Business Logic
- Status Management
- Query Optimization

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Week 07 (31 Aug – 04 Sept 2026): Clinic Appointment System

Tables

- Patients
- Doctors
- Appointments

Features

- Schedule appointments
- Doctor availability
- Appointment history

Concepts

- Lookup Tables
- Validation Rules
- Advanced Relationships

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Week 08 (07 – 11 Sept 2026): Inventory Management System

Tables

- Products
- Categories
- Suppliers
- Stock Movements

Features

- Stock control
- Reordering alerts
- Supplier management

Concepts

- Aggregation Queries
- Inventory Calculations
- Reporting

C# Concepts

- Dashboards
- Stock Reports
- Alerts

NB: The Summative Assessment for MDB622 is scheduled for the 17th of September 2026, so use the following week for revision instead

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Week U: Point of Sale (POS) System

Tables

- Products
- Customers
- Sales
- SaleDetails

Features

- Process sales
- Generate receipts
- Daily sales reports

Concepts

- Transactions
- Master-Detail Relationships
- Stored Procedures

SQL Skills

SQL

1 BEGIN TRANSACTION

2 COMMIT

3 ROLLBACK

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Week V: School Registration System

Tables

- Students
- Modules
- Registrations

Features

- Student registration
- Module enrollment
- Academic records

Concepts

- Many-to-Many Relationships
- Junction Tables

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PHASE 3: ADVANCED DATABASE DEVELOPMENT

Week W: Asset Tracking System

Tables

- Assets
- Employees
- AssetAssignments

Features

- Asset allocation
- Asset returns
- Audit trail

Concepts

- Auditing
- Asset Lifecycle Management
- Reporting

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Week X: Help Desk / Ticketing System

Tables

- Users
- Tickets
- Categories
- Statuses

Features

- Log tickets
- Assign technicians
- Track ticket progress

Concepts

- Workflow Management
- Status Tracking
- User Roles

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Week Y: Pharmacy Management System

Tables

- Medicines
- Suppliers
- Customers
- Sales

Features

- Medicine tracking
- Expiry-date monitoring
- Sales processing

Concepts

- Inventory Management
- Alerts
- Reporting

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Week Z: User Authentication Project

Tables

- Users
- Roles
- Permissions

Features

- Login
- Role-based access
- Password hashing

Concepts

- Authentication
- Authorization
- Security

SQL Topics

- Roles
- Permissions
- Database Security

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PHASE 4: CAPSTONE PROJECTS

Students select ONE project.

Option 1 – Learning Management System (LMS)

Tables

- Students
- Courses
- Assessments
- Enrolments

Features

- Student enrollment
- Marks capture
- Transcripts

Skills Covered

- CRUD
- Relationships
- Reporting
- Authentication

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Option 2 – Payroll System

Tables

- Employees
- Payroll
- Deductions
- Allowances

Features

- Salary calculation
- Payslip generation

Skills Covered

- Stored Procedures
- Calculations
- Reports

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Option 3 – E-Commerce System

Tables

- Customers
- Products
- Orders
- OrderDetails

Features

- Shopping cart
- Order processing
- Sales reports

Skills Covered

- Transactions
- Inventory Updates
- Reporting

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Option 4 – Hostel Residence Management System

Tables

- Students
- Rooms
- Allocations

Features

- Room allocation
- Occupancy reports
- Student tracking

Option 5 – Examination Management System

Tables

- Students
- Subjects
- Exams
- Results

Features

- Mark capture
- Average calculations
- Report generation

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Additional Assessment Requirements

Each project must include:

Database Requirements

- Minimum 4 tables
- Proper PK/FK relationships
- Constraints
- Normalization to 3NF

SQL Requirements

- Minimum 10 INSERT statements
- Minimum 5 UPDATE statements
- Minimum 5 DELETE statements
- Minimum 5 JOIN queries
- Minimum 2 Views
- Minimum 2 Stored Procedures

C# Requirements

- Login screen
- Dashboard
- Add/Edit/Delete forms
- Search functionality
- DataGridView integration
- Exception handling

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Documentation Requirements

- Problem Statement
- ERD
- Data Dictionary
- Screenshots
- SQL Script
- User Manual

Recommended Software Stack

- Microsoft SQL Server 2022
- SQL Server Management Studio (SSMS)
- Visual Studio 2022
- C# Windows Forms
- ADO.NET
- GitHub (Version Control)

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